

1.1 a. M_1, M_2 的起始状态都是 q_1

b. 接受状态: $M_1: q_2, M_2: q_1, q_4$

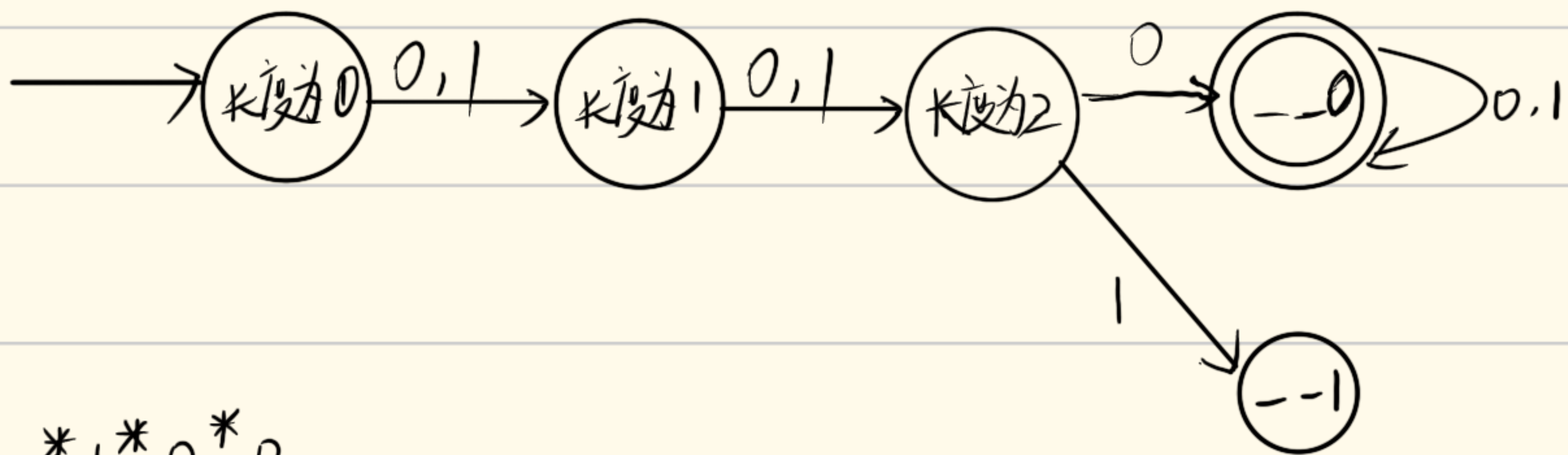
c. $M_1: q_1 \xrightarrow{a} q_2 \xrightarrow{a} q_3 \xrightarrow{b} q_1 \xrightarrow{b} q_1$

$M_2: q_1 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{b} q_4$

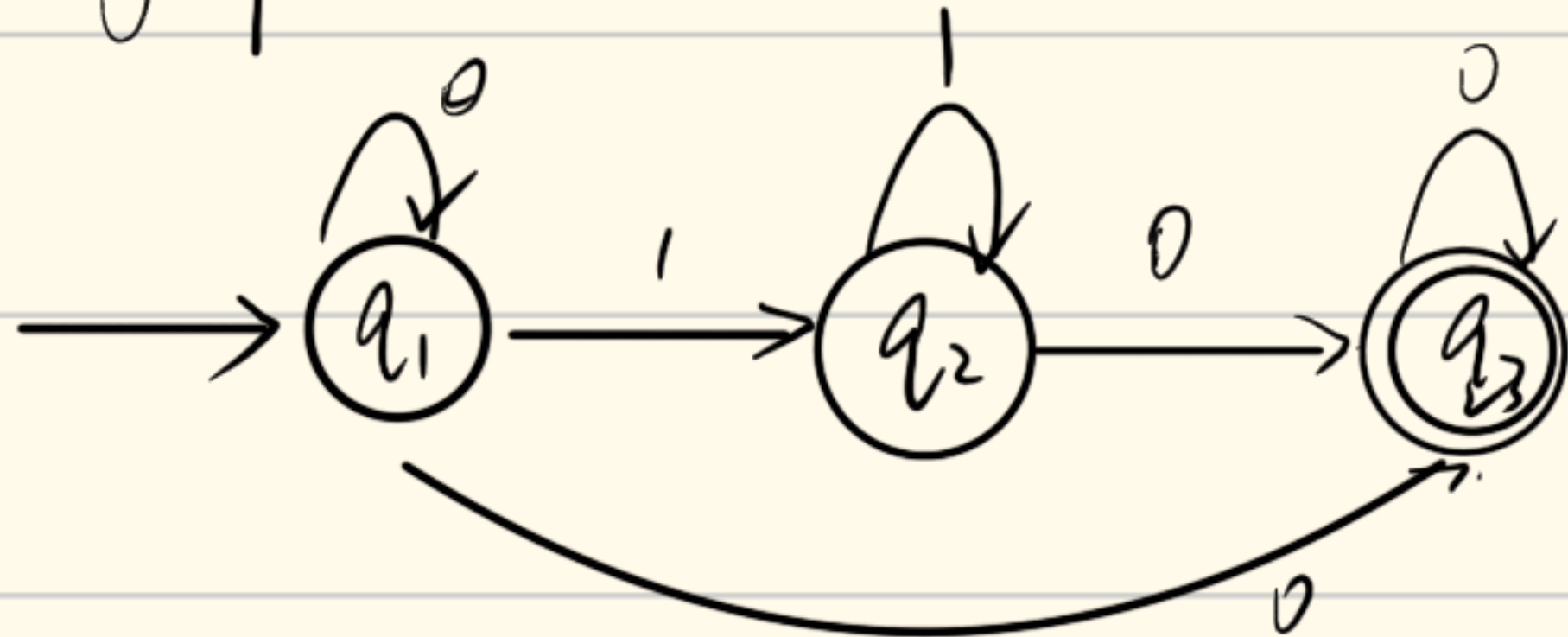
d. M_1 拒绝, M_2 接受 $aabb$.

e. M_1 拒绝 ϵ , M_2 接受 ϵ .

1.6.

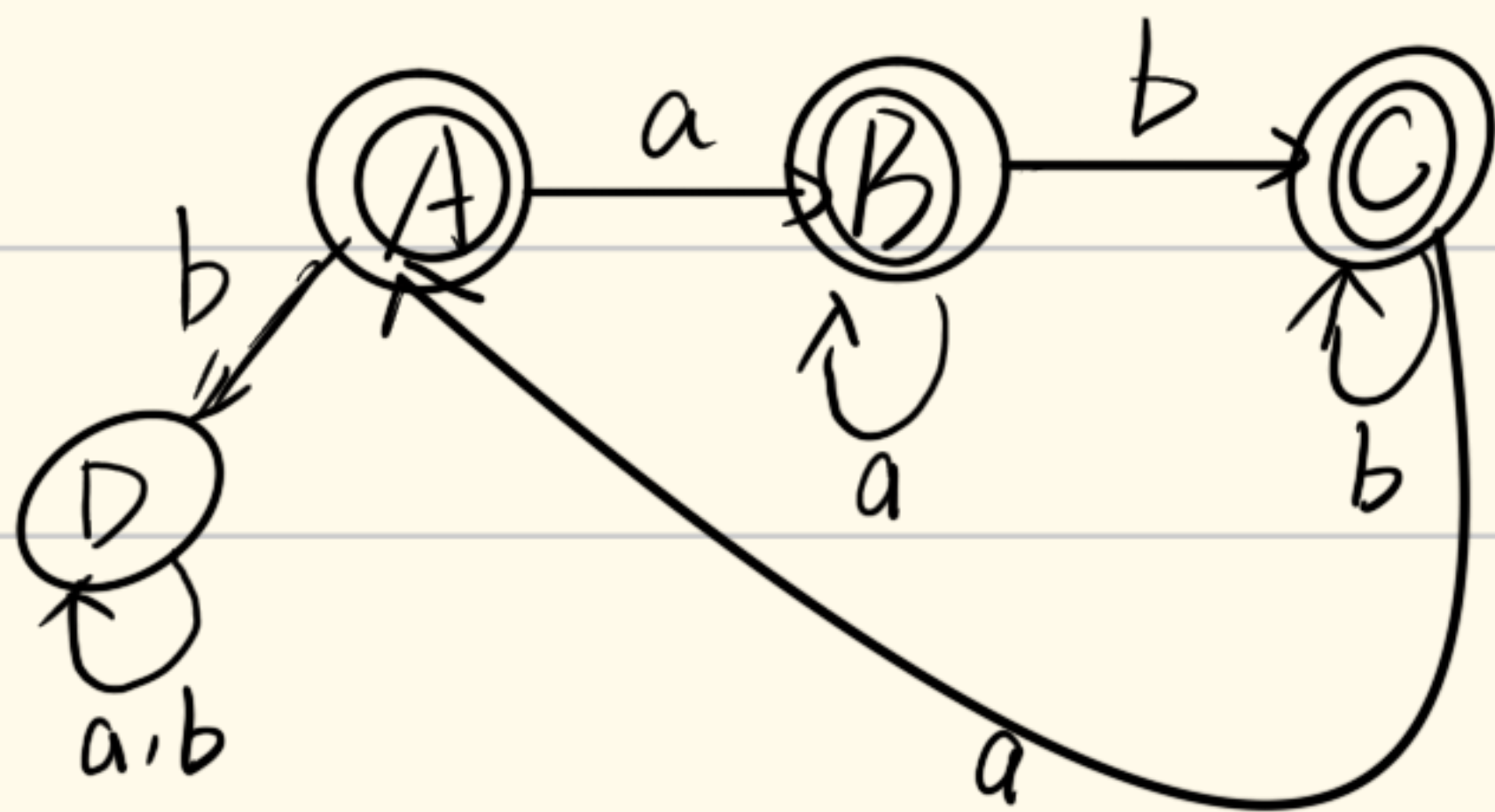


1.7 $0^*1^*0^*0$



1.16. $\{1,2\} \xrightarrow{a} \{\}, \{1,2\} \xrightarrow{b} \{3,2\} \xrightarrow{a} \{2,1\}$
 $\downarrow b \quad \quad \quad \uparrow a \quad \quad \quad \downarrow b$
 $\emptyset \quad \quad \quad a \quad \quad \quad b$

$A = \{1,2\}, B = \{1,2,3\}, C = \{2,3\}, D = \emptyset$

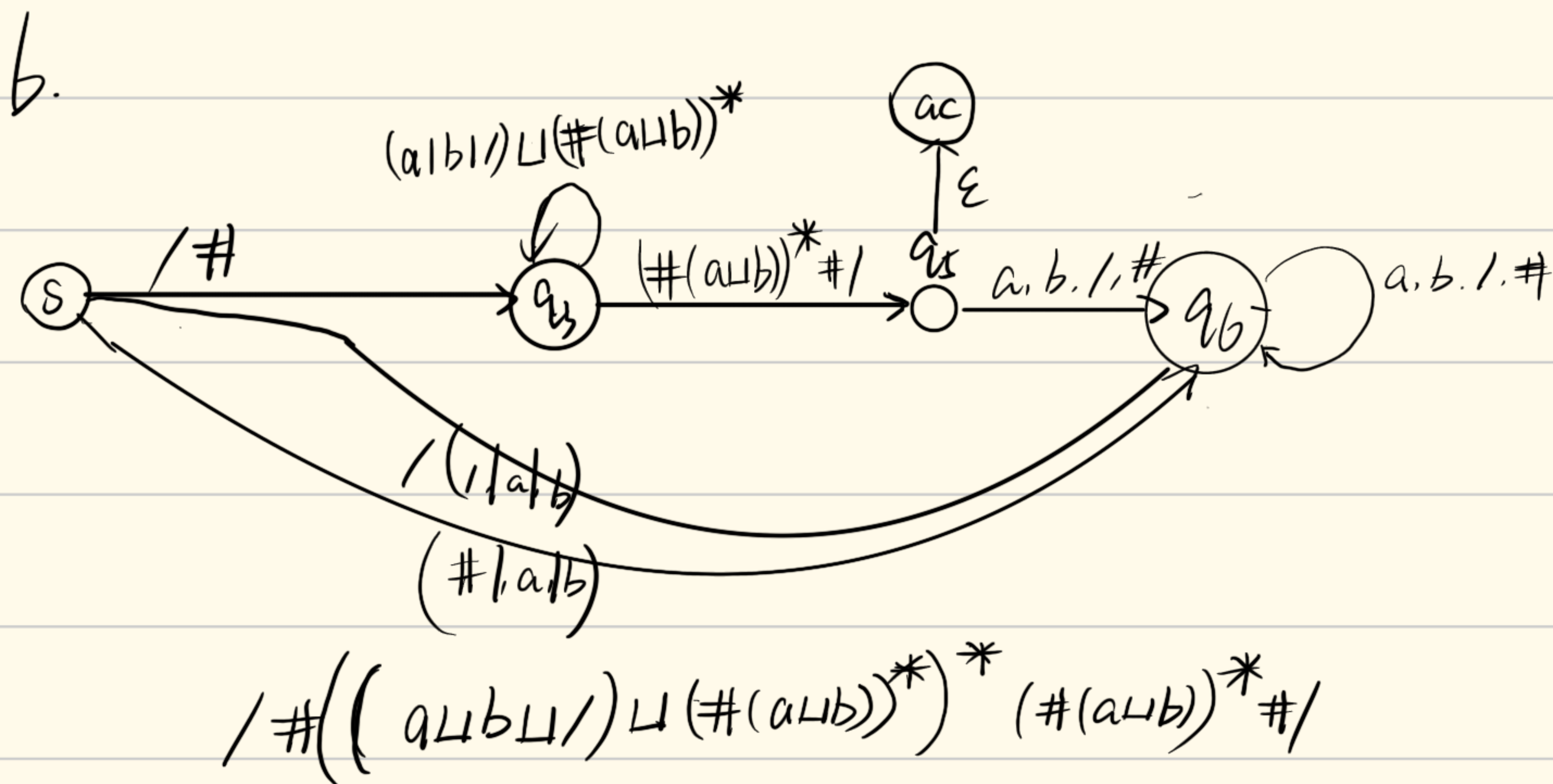
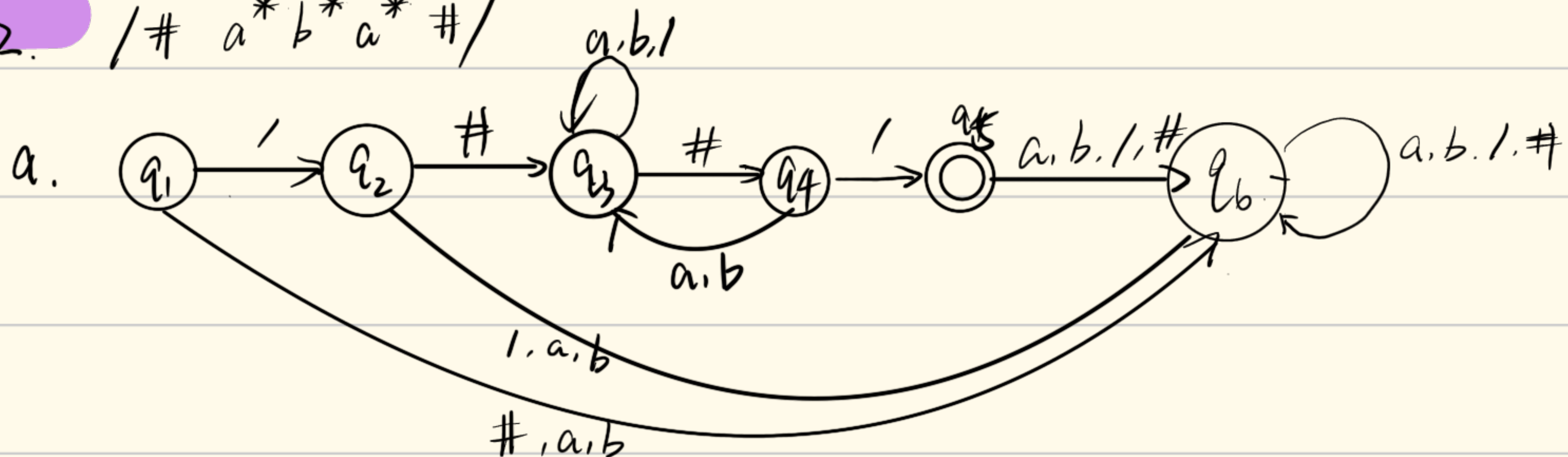


A, B, C 都是接受态.

1.21 含奇数个 b.

$a^*b(a^*ba^*b)^*a^*$

1.22. $/\# a^* b^* a^* \# /$



1.29. 设 $A = \{www \mid w \in \{a, b\}^*\}$ 正则

令 $w = a^p \cdot b$, $|xy| \leq p$, $|y| > 0$, $\therefore x, y$ 都在第1个 w 的 a^p 里

设 $y = a^k$, $0 < k < p$, 取 $i = 0$, $xy^0z = (a^{p-k}b)ww$.

很明显此时 $xy^0z \notin A$.

$\therefore A$ 不是正则语言.